Shicheng PENG

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Research Interests

I study robot perception under **ambiguous partial observations**, particularly for deformable objects, understanding why learned models fail, and using interaction to resolve ambiguity.

Education

University of Michigan, Ann Arbor Aug. 2024 – May 2027 (expected)
M.S. in Robotics & M.S. in Applied Statistics (Dual Degree) Ann Arbor, MI, USA
GPA: 4.0/4.0

The Chinese University of Hong Kong, Shenzhen Sept. 2020 – May 2024
B.S. in Financial Engineering Shenzhen, China
GPA: 3.8/4.0 (Top 5%)

Publications and Manuscripts

Chen, Y., Sang, F.*, **Peng, S.***, Wu, X.*, Zong, Y., Chen, Y., Li, A., Zhang, B., & Vasudevan, R. (2026). *DEFT: Differentiable Branched Discrete Elastic Rods for Modeling Furcated DLOs in Real-Time*. Accepted by *The International Journal of Robotics Research*.
* Equal contribution.

Peng, S. (2023). *Modeling the pork price cycle in China based on the age structure of hogs. Frontiers in Artificial Intelligence and Applications*, 373, 995–1005. DOI.

Research Experience

ROAHM Lab Nov. 2025 – Present
Research Assistant; Advisor: Prof. Ram Vasudevan University of Michigan

Ambiguity-Aware Garment Perception for Active Manipulation Sept. 2026 – Present

- **Led the formulation of an ambiguity-aware garment perception project**, building on an initial idea from my PI and mentor and aiming to test whether uncertainty signals from single-view observations can guide active perception toward accurate garment mesh reconstruction with fewer interactions. Established baseline evaluations of geometric accuracy and correspondence.
- **Diagnosed symmetry-induced mean collapse in square-cloth reconstruction**. Traced the prediction area collapse under L_2 supervision to a conflict between rotationally indistinguishable observations and fixed template material labels. Devised a topology-preserving spatial ordering to resolve this conflict, reducing validation vertex error from **254.2 mm to 12.8 mm** using the same data and training settings.
- **Uncovered persistent mirror-correspondence errors in UniClothDiff** despite approximately **3 mm surface error**. Probed cross-attention through targeted input interventions and identified its dependence on the noisy mesh estimate. Proposed a hierarchical correspondence design that separates hidden-shape completion from fine-grained matching to observed points.

DiffADMM: Differentiable Cloth Simulation May 2026 – Present

- **Accelerated cloth bending computation through GPU kernel redesign**. Used matched-topology rope–cloth comparisons and runtime scaling analysis to pinpoint uneven workloads in vertex-parallel bending computation as the bottleneck. Reorganized the kernel around bending elements, eliminating the observed runtime plateau and achieving a **3.00×–6.74× speedup in bending computation**.
- **Derived the missing gradient path for pinned-boundary constraints**. Traced configuration-dependent gradient failures to inconsistent boundary treatment between the forward and backward passes, and corrected the backward formulation. Verified the corrected gradients using finite differences and rope–cloth consistency checks, with maximum relative error below 10^{-3} across the tested gradients and configurations.
- **Investigating self-contact integration** into the differentiable ADMM cloth simulator, focusing on nonpenetration constraints and their compatibility with the existing solver and gradient computation.

Research Experience (continued) — ROAHM Lab

DEFT: Modeling and Simulation of Branched Deformable Linear Objects Nov. 2025 – May 2026

- **Co-designed and implemented DEFT's simulation evaluation**. Established the shape-matching evaluation protocol, trained baseline models, and conducted comparisons and ablations.
- **Built the inverse-kinematics solver** connecting simulated manipulation trajectories to Kinova Gen3 execution. Attributed convergence failures in IK optimization to discontinuities in the orientation-error representation, and reformulated the residual using an $SE(3)$ log map. Further isolated the remaining failures to robot-model mismatches in the URDF and corrected them to recover physically executable trajectories.
- **Jointly designed and conducted real-robot evaluations** of parent-branch and child-branch wire insertion. Utilized motion capture to prepare reproducible BDLO configurations and target poses. Refined simulation scripts to correct the optimization objective for high-level insertion waypoints, and evaluated DEFT against baseline methods.

Selected Academic Projects

Influence-Based Data Selection for Object Detection Mar. 2026 – Apr. 2026

Project Lead & Developer University of Michigan

- **Developed sample selection using influence functions** to adapt a pretrained YOLO detector effectively with less training data and fewer epochs. Ranked a large labeled target-domain pool by each sample's influence score, computed on a separate labeled probe set using an empirical Fisher approximation and conjugate gradients. Fine-tuned the detection head on 100 selected images per condition.
- **Compared the method with three selection baselines** across four adverse driving conditions using the same candidate pool and fine-tuning budgets, with test data kept separate from the probe set. The largest improvements occurred in nighttime and snowy conditions, suggesting greater benefit under larger domain shifts.
- **Tested sensitivity to data and training budgets** by varying probe-set size, selected-sample count, and fine-tuning epochs for nighttime detection. Also deployed the implementation on a Raspberry Pi 5–based MBot and verified CPU-only execution, confirming operation on a compute-constrained edge device.

Point Cloud Registration with Modified Feature Histograms Nov. 2025 – Dec. 2025

Algorithm Implementation & Experiment Lead University of Michigan

- **Modified Point Feature Histogram (PFH) construction** to use only query–neighbor pairs, reducing computation complexity while retaining informative features for alignment. Evaluated accuracy–runtime trade-offs across six real-world point cloud datasets, varying descriptors, histogram binning, and correspondence metrics.
- **Tuned voxel resolution and density filtering**, saving **93–98%** of preprocessing and feature-matching time on test point clouds relative to configurations without filtering, without reducing registration success rate.

Honors & Awards

The Chinese University of Hong Kong, Shenzhen

Tier 1 Academic Performance Scholarship, highest annual GPA in cohort 2023

Dean's List, awarded three times 2021–2024

Programming Contest, Third Prize 2024

Teaching Experience

The Chinese University of Hong Kong, Shenzhen

CSC 3100 Data Structures, Peer Tutor (selected based on programming contest performance) 2024

FIN 4120 Fixed Income Securities Analysis, Undergraduate Student Teaching Staff 2023

Technical Skills

Programming: Python, C++, R, MATLAB, Bash
Robotics & ML: PyTorch, Pinocchio, Open3D, Isaac Sim
Development Tools: Linux, Git, Docker

Selected Coursework

ML & Robotics: Deep Learning for Robot Perception, Algorithmic Robotics, Machine Learning
Statistics & Probability: Bayesian Modeling, Stochastic Processes, Causal Inference